

Can there be only one?

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Hirschian “quantifier variance”

1. There are several possible languages, syntactically alike, which differ as regards the meanings of quantifier-words (inter alia).
2. Many of the sentences ontologists debate have different truth values in the different languages.
3. The same range of Kaplanian “characters” is expressible in all the languages.

To the extent that one can associate a sentence's character with its "meaning" the stipulation is that in shifting from one language to another nothing is gained or lost in what can be "meant" by one's asserted sentences. [235]

4. The “formal logical properties” of the quantifier-words are the same in all the languages.

[T]he imagined change in the meaning of the expression "there exists something" will leave the expression's general role in the language largely intact. In particular, the purely syntactic and formal logical properties of the expression will not be changed at all (the formal principles of quantificational logic will be unaltered)

Why does it matter? The argument from charity

1. *Principle of charity*: Other things being equal, if it is possible to interpret a community as speaking a language in which the sentences that strike speakers as obvious come out true, it is correct to do so.
2. It is possible to interpret our community as speaking a language in which the sentences [in class C] that strike us as obvious come out true. (*By quantifier variance*)
3. Other things are equal.
4. So the sentences [in C] that strike us as obvious are true in our language.
5. ‘Nothing in my garden keeps gaining and losing branches’ strikes us as obvious [and is in C].
6. So it is true in our language.
7. So nothing in my garden keeps gaining and losing branches.

Harris’s theorem

‘If we have two systems of candidates for what the logical operators mean, both satisfying the rules of inference, then a sentence containing operators from one set and the corresponding sentence with operators from the other set are interderivable....[T]he theorem assures us that the rules succeed in pinning down the roles of the logical operations uniquely to within logical equivalence.’ [McGee]

The case of disjunction

Inferential role premises:

$\vee I$: $[\phi] \vdash [\phi \vee \psi]$ and $[\psi] \vdash [\phi \vee \psi]$.

$\vee E$: Whenever $[\phi] \vdash [\chi]$ and $[\psi] \vdash [\chi]$, $[\phi \vee \psi] \vdash [\chi]$.

Translation premises

T1 (a) $[\phi]_1 \Leftrightarrow [\phi']_2$

(b) $[\psi]_1 \Leftrightarrow [\psi']_2$

T2 (a) $[\phi \vee \psi]_1 \Leftrightarrow [\chi']_2$

(b) $[\phi' \vee \psi']_2 \Leftrightarrow [\chi]_1$

Argument:

(1) $[\phi]_1 \vdash [\phi' \vee \psi']_2$ and $[\psi]_1 \vdash [\phi' \vee \psi']_2$ by $\vee I_2$ and T1

(2) $[\phi \vee \psi]_1 \vdash [\phi' \vee \psi']_2$ by $\vee E_1$ and T2(a)

and conversely,

(3) $[\phi']_2 \vdash [\phi \vee \psi]_1$ and $[\psi']_2 \vdash [\phi \vee \psi]_1$ by $\vee I_1$ and T1

(4) $[\phi' \vee \psi']_2 \vdash [\phi \vee \psi]_1$ by $\vee E_2$ and T2(b)

Questions about the disjunction case

- (A) How do we get from use to the inferential role premises?
- (B) Are we justified in taking the conclusion to mean that there is no important difference in the meaning of $\vee$ between L_1 and L_2 ?

The case of disjunction

Inferential role premises:

$\vee I$: $[\phi] \vdash [\phi \vee \psi]$ and $[\psi] \vdash [\phi \vee \psi]$.

$\vee E$: Whenever $[\phi] \vdash [\chi]$ and $[\psi] \vdash [\chi]$, $[\phi \vee \psi] \vdash [\chi]$.

$\vee E+$: Whenever $[\phi] \vdash p$ and $[\psi] \vdash p$, $[\phi \vee \psi] \vdash p$.

?

Translation premises

T1 (a) $[\phi]_1 \Leftrightarrow [\phi']_2$

(b) $[\psi]_1 \Leftrightarrow [\psi']_2$

T2 (a) $[\phi \vee \psi]_1 \Leftrightarrow [\chi']_2$

(b) $[\phi' \vee \psi']_2 \Leftrightarrow [\chi]_1$

Argument:

(1) $[\phi]_1 \vdash [\phi' \vee \psi']_2$ and $[\psi]_1 \vdash [\phi' \vee \psi']_2$ by $\vee I_2$ and T1

(2) $[\phi \vee \psi]_1 \vdash [\phi' \vee \psi']_2$ by $\vee E_1$ and T2(a)

and conversely,

(3) $[\phi']_2 \vdash [\phi \vee \psi]_1$ and $[\psi']_2 \vdash [\phi \vee \psi]_1$ by $\vee I_1$ and T1

(4) $[\phi' \vee \psi']_2 \vdash [\phi \vee \psi]_1$ by $\vee E_2$ and T2(b)

Quantification with names

Inferential role premises:

$\exists I$: $[\phi(c)] \vdash [\exists v \phi(v)]$

$\exists E$: Whenever $[\phi(c)] \vdash [\chi]$ for some c -free sentence χ , $[\exists v \phi(v)] \vdash [\chi]$.

Translation premises:

T1 $[\phi(c)]_1 \Leftrightarrow [\phi'(d)]_2$

for some c, d

T2 (a) $[\exists x \phi(x)]_1 \Leftrightarrow [\chi']_2$

(b) $[\exists y \phi(y)]_2 \Leftrightarrow [\chi]_1$

for c -free χ and d -free χ'

Argument:

(1) $[\phi(c)]_1 \vdash [\exists y \phi'(y)]_2$ by $\exists I_2$ and T1

(2) $[\exists x \phi(x)]_1 \vdash [\exists y \phi'(y)]_2$ by $\exists E_1$ and T2(a)

and conversely,

(3) $[\phi(d)]_2 \vdash [\exists x \phi'(x)]_1$ by $\exists I_1$ and T1

(4) $[\exists y \phi'(y)]_2 \vdash [\exists x \phi(x)]_1$ by $\exists E_2$ and T2(b)

New worries about this argument

(C) Problems for $\exists E$:

- (i) [John is from Germany] $\vdash$ [John is German] but
[John is from something] $\nvdash$ [John is German]
- (ii) [Mike is a hammer] $\vdash$ [Head is attached to Handle]
but [Something is a hammer] $\nvdash$ [Head is attached to Handle].
 - *Response*: ' $\vdash$ ' stands for *narrowly logical* entailment.
 - *Rejoinder*: even if it makes sense to apply such a notion to meanings, the demand that every sentence in L_1 be *narrowly logically equivalent* to some sentence of L_2 is excessively strong.

(D) Problems for $\exists I$:

- (i) *Empty names*: [Vulcan is nonexistent] $\nvdash$ [Something is nonexistent].
- (ii) *Names for contingent existents*: [John is nonexistent] $\nvdash$ [Something is nonexistent].

(E) A natural thought: differences in the meanings of quantifiers will correlate with difference in the meanings of names.

- If so, a subject-predicate sentence $\phi(c)$ in L_1 might not be equivalent to any subject-predicate sentence $\phi'(d)$ of L_2 .

Quantification with free variables

Inferential role premises:

$\exists I$: $[\phi(v)] \vdash [\exists v\phi(v)]$

$\exists E$: Whenever $[\phi(v)] \vdash p$ and $p = [\chi]$ for some v -free χ , $[\exists v\phi(v)] \vdash p$.

Translation premises

T1 $[\phi(x)]_1 \Leftrightarrow [\phi'(y)]_2$
for some x, y

T2 (a) $[\exists x\phi(x)]_1 \Leftrightarrow [\chi']_2$
(b) $[\exists y\phi(y)]_2 \Leftrightarrow [\chi]_1$
for x -free χ and y -free χ'

Argument:

(1) $[\phi(x)]_1 \vdash [\exists y\phi'(y)]_2$	by $\exists I_2$ and T1
(2) $[\exists x\phi(x)]_1 \vdash [\exists y\phi'(y)]_2$	by $\vee E_1$ and T2(a)

...and analogously:

(1) $[\phi(y)]_2 \vdash [\exists x\phi'(x)]_1$	by $\exists I_1$ and T1
(2) $[\exists y\phi'(y)]_2 \vdash [\exists x\phi(x)]_1$	by $\vee E_2$ and T2(b)

Problems with open sentences

What sort of thing is $[\phi]$ when ϕ contains free variables?

- How is $[\phi(x)]$ different from $[\phi(y)]$?
- And given that there must be a difference in these cases, what could ever make the meaning of an open sentence in one language equivalent to the meaning of an open sentence in a different language?

Solution

Work with *predicates* instead of open sentences

- $[\lambda x\phi(x)] = [\lambda y\phi(y)]$
- Write $^x[\phi(x)]$ for $[\lambda x\phi(x)]$

Quantification with predicates

Inferential role premises:

$\exists I$: $^v[\phi(v)] \vdash ^v[\exists v\phi(v)]$
 $\exists E$: Whenever $^v[\phi(v)] \vdash ^v[\chi]$ for some closed χ , $[\exists v\phi(v)] \vdash [\chi]$.

Translation premises

T1 $^x[\phi(x)]_1 \Leftrightarrow ^y[\phi'(y)]_2$
 for some x, y
 T2 (a) $[\exists x\phi(x)]_1 \Leftrightarrow [\chi']_2$
 (b) $[\exists y\phi(y)]_2 \Leftrightarrow [\chi]_1$
 for closed χ and closed χ'

Argument:

(1) $^x[\phi(x)]_1 \vdash ^y[\exists y\phi'(y)]_2$ by $\exists I_2$ and T1
 (2) $[\exists x\phi(x)]_1 \vdash [\exists y\phi'(y)]_2$ by $\exists E_1$ and T2(a)

and conversely,

(3) $^y[\phi(y)]_2 \vdash ^x[\exists x\phi'(x)]_1$ by $\exists I_1$ and T1
 (4) $[\exists y\phi'(y)]_2 \vdash [\exists x\phi(x)]_1$ by $\exists E_2$ and T2(b)

Quantification with ~~free~~ predicates

Inferential role premises:

$\exists I$: $^v[\phi(v)] \vdash ^v[\exists v\phi(v)]$
 $\exists E$: Whenever $^v[\phi(v)] \vdash ^v[\chi]$ for some closed χ , $[\exists v\phi(v)] \vdash [\chi]$.
 $\exists E+$: Whenever $^v[\phi(v)] \vdash q$ and q is the unary expansion of p , $[\exists v\phi(v)] \vdash p$

Translation premises

T1 $^x[\phi(x)]_1 \Leftrightarrow ^y[\phi'(y)]_2$
 for some x, y
 T2 (a) $[\exists x\phi(x)]_1 \Leftrightarrow [\chi']_2$
 (b) $[\exists y\phi(y)]_2 \Leftrightarrow [\chi]_1$
 for closed χ and closed χ'

?

Argument:

(1) $^x[\phi(x)]_1 \vdash ^y[\exists y\phi'(y)]_2$ by $\exists I_2$ and T1
 (2) $[\exists x\phi(x)]_1 \vdash [\exists y\phi'(y)]_2$ by $\exists E_1$ and T2(a)

and conversely,

(3) $^y[\phi(y)]_2 \vdash ^x[\exists x\phi'(x)]_1$ by $\exists I_1$ and T1
 (4) $[\exists y\phi'(y)]_2 \vdash [\exists x\phi(x)]_1$ by $\exists E_2$ and T2(b)

How this plays out in practice

- We're tempted to translate PVI's "something" as "something simple or living", and DKL's "something..." as "if composition were universal, something...".
- But if so, $\exists I$ would fail in PVIish: $[is\ heavy]_{PVI}$ would not entail $[is\ such\ that\ something\ is\ heavy]_{PVI}$.
- And $\exists E$ would fail in DKLish: although $[is\ heavy]_{DKL}$ does entail $[is\ such\ that\ something\ is\ heavy]_{DKL}$, $[Something\ is\ heavy]_{DKL}$ doesn't entail $[Something\ is\ heavy]_{English}$.

Central worries about this argument

- (A) Even if we can get from use to conclusions about entailment among *sentences*, how do we get from use to conclusions about entailment among *predicates*?
- (B) Even if we have reason to think that the meanings expressible by sentences of L_1 are the same as those expressible by sentences of L_2 , up to equivalence, do we have reason to think the same is true for predicates?

Two ways of thinking about equivalence among properties:

- (i) ' $F \Leftrightarrow G$ ' means something like: Necessarily, for any x , necessarily, Fx iff Gx .
- (ii) ' $F \Leftrightarrow G$ ' just means *to be F is to be G*; quantifiers are not involved.

Hirsch's view: the meanings of sentences are explanatorily prior to the meanings of non-sentences.

- 'the essence of language is the distribution of a set of characters over a set of syntactically structured sentences'.
- What must be given up is a picture of language in which the characters at the level of sentences are generated by some underlying referential mechanisms at the level of words. This "bottom-up" picture is misguided because the references of words depend upon the characters of sentences.

An example

‘Something...’ in 4Dish means what ‘If the doctrine of temporal parts were true, something...’ means in English.

- Case 1: the DTP is contingent. Then 4Dish will be impoverished: none of its sentences is equivalent to the English ‘There are temporal parts’.
- Case 2: the DTP is necessarily false (equivalent to a contradiction). Then, plausibly, [temporal part] $\Leftrightarrow$ [nonselfidentical thing].

Making things work in free logic (1)

Inferential role premises:

$\forall E$: $*[(\forall y\phi(y) \wedge \exists y(y=x)) \vdash *[\phi(x)]]$

Id: If ${}^{uv}[\psi(u) \wedge \phi(v)] \vdash {}^{uv}[u=v]$, where v is not free in ψ and u is not free in ϕ , then ${}^v[\phi(v)] \vdash {}^v[\exists u(u=v)]$

Translation premises

- T (a) ${}^x[\exists y(y=x)]_1 \Leftrightarrow {}^x[\phi'(x)]_2$
 (b) ${}^x[\forall y(y=x)]_1 \Leftrightarrow {}^x[\psi'(x)]_2$
 (c) ${}^x[\exists y(y=x)]_2 \Leftrightarrow [\phi(x)]_1$
 (d) ${}^x[\forall y(y=x)]_2 \Leftrightarrow [\psi(x)]_1$

Argument:

- (1) ${}^{xy}[x=y]_1 \Leftrightarrow {}^{xy}[x=y]_2$ Harris arg. for ‘=’
 (2) ${}^{xy}[\forall z(z=y) \wedge \exists z(z=x)]_1 \Leftrightarrow {}^{xy}[\psi'(y) \wedge \phi'(x)]_2$ T(a-b), Harris arg. for $\wedge$
 (3) ${}^{xy}[\forall z(z=y) \wedge \exists z(z=x)]_1 \vdash {}^{xy}[x=y]_1$ free $\forall E$
 (4) ${}^{xy}[\psi'(y) \wedge \phi'(x)]_2 \vdash {}^{xy}[x=y]_2$ (1-3)
 (5) ${}^x[\phi'(x)]_2 \vdash {}^x[\exists y(x=y)]_2$ Id
 (6) ${}^x[\exists y(y=x)]_1 \vdash {}^x[\exists y(x=y)]_2$ T(a)
 and analogously, ${}^x[\exists y(y=x)]_2 \vdash {}^x[\exists y(x=y)]_1$

Making things work in free logic (2)

Inferential role premises:

Free $\exists I$:

${}^v[\phi(v) \wedge \exists u(u=v)] \vdash {}^v[\exists v\phi(v)]$

$\exists E$: Whenever ${}^v[\phi(v)] \vdash {}^v[\chi]$ for some closed χ , $[\exists v\phi(v)] \vdash [\chi]$.

Translation premises

T1 ${}^x[\phi(x)]_1 \Leftrightarrow {}^y[\phi'(y)]_2$

for some x, y

T2 (a) $[\exists x\phi(x)]_1 \Leftrightarrow [\chi']_2$

(b) $[\exists y\phi(y)]_2 \Leftrightarrow [\chi]_1$

for x -free χ and y -free χ'

Argument:

(1) ${}^x[\phi(x) \wedge \exists z(z=x)]_1 \Leftrightarrow {}^x[\phi'(x) \wedge \exists z(z=x)]_2$

by T1, the previous result and Harris for $\wedge$

(2) ${}^x[\phi(x) \wedge \exists z(z=x)]_1 \vdash {}^y[\exists y\phi'(y)]_2$ by Free $\exists I_2$

(3) $[\exists x(\phi(x) \wedge \exists z(z=x))]_1 \vdash [\exists y\phi'(y)]_2$ by $\exists E_1$ and T2(a)

(4) $[\exists x\phi(x)]_1 \vdash [\exists y\phi'(y)]_2$ $\wedge$ -Elim

and analogously, $[\exists y\phi'(y)]_2 \vdash [\exists x\phi(x)]_1$